

This country profile was compiled by the OECD Secretariat and reflects information available as of June 2013. Further information and analysis can be found in the publication: OECD (2013) *Water and Climate Change Adaptation: Policies to Navigate Uncharted Waters*, OECD Studies on Water, OECD Publishing. <http://dx.doi.org/10.1787/9789264200449-en>. Country profiles for all OECD member countries are available for download at: www.oecd.org/env/resources/waterandclimatechange.htm. These profiles will be regularly updated and it is planned to expand coverage over time to include key partner countries.

Luxembourg

Climate change impacts on water systems

Observed changes and trends	• Increase in annual mean temperatures over the last decades. The annual mean temperature for the capital that was typical for the second half of the 20th century (around 9 °C) is now regularly exceeded. Since 2000, the annual mean temperature is between 9.3 °C (2001) and 11.3 °C (2007). This rise is mainly driven by higher temperatures in winter. • Seasonal distribution of precipitation has shown substantial variability over the past 130 years. • Increased frequency of floods due to major redistributions of winter rainfall has caused an increase in maximum daily runoff during winter. • Increase in the frequency and length of dry periods.				
Projected impacts	• Increase in temperature from 1 °C to 2.2 °C by 2050, as compared to the period 1960-90. Annual mean temperature is expected to reach up to 11.6 °C for the period 2071 to 2100. • Relatively stable annual totals of precipitation up to 2100 suggested by preliminary studies. However, a substantial redistribution of seasonal precipitation totals can be expected in the second half of the century, with a decrease in summer rainfall and an increase in winter precipitation. In winter, increase in rainfall from 0 to 25% by 2050 with increasing discharges. In summer, decrease in rainfall of 5 to 25% with reduced run-off by 2050. • Deterioration of water quality due to intensification of rainfalls (increasing erosions, rapid infiltration towards groundwater) and shift in the main recharge period of groundwater. • Changes in the water cycle could increase public health risks related to water quality and water scarcity. • Increase in the risk of river flood in winter and an increase in the risk of droughts.				
Primary concerns	Water quantity	Water quality	Water supply and sanitation	Extreme weather events	Ecosystems
	✓	✓	✓	✓ (floods)	
Key vulnerabilities	• Drinking water supply due to an expected increase in drinking water consumption as a result of demographic changes.¹ • Groundwater recharge due to a shift of the main recharge period. • Changes in the hydrological regime of rivers. • Water quality deterioration due to an intensification of rainfall, increasing erosion, rapid infiltration of groundwater, etc. • Flood and low water management. • Navigation.				

1. Possible restrictive water use measures to be put in place during warm periods in order to manage water consumption peaks. Drinking water supply also has to deal with 150 000 daily commuters from bordering countries.

Sources: Department of Environment (2010), *Fifth National Communication of Luxembourg*, http://unfccc.int/national_reports/annex_i_natcom/submitted_natcom/items/4903.php (accessed 20 June 2012); Görgen, K. et al. (2010), *Assessment of Climate Change Impacts on Discharge in the Rhine River Basin: Results of the RheinBlick2050 Project*, CHR report, I-23, 229 pp., Lelystad, ISBN 978-90-70980-35-1; International Commission for the Protection of the Rhine (2011), *Study of Scenarios for the Discharge Regime of the Rhine*, www.iksr.org/fileadmin/user_upload/Dokumente_en/Reports/188_e.pdf (accessed 12 August 2012).

Key policy documents

Document	Reference to water?	Type	Year	Responsible institution
National Adaptation Plan ¹	Y	National adaptation plan	2011	

1. Adopted in June 2011 by Luxembourg's Council of Ministers, the plan prioritises biodiversity, water, agriculture and forestry.

Policy instruments

Areas	Policy mix	Regulatory instruments	Economic instruments	Information and other instruments
Water quantity		- Licensing and metering of surface and groundwater abstractions. - Prohibition of certain water uses (e.g. irrigation) in order to guarantee water supply during critical periods (e.g. "Phase orange", "Phase rouge").		- Efforts to reduce the inefficient water use and promote water saving infrastructures and rainwater catchment systems. - Campaigns to raise awareness about saving drinking water. - Protection against erosion, development of sustainable soil use. - Preliminary flood risk assessment has been undertaken along with the preparation of flood hazard maps and flood risk maps. The establishment of a flood risk management plan is planned for 2015. - Early warning systems: Luxembourg has a dense national network of meteorological stations and hydrological stations and initiatives for the public dissemination of information (competent authorities and the public are informed and warned about floods ± 48 hours before the flood event.¹ - Additional natural retention volume to promote prevention of flash floods. - Efforts to avoid additional stress on water resources that may result from new surface and groundwater water pollution sources due to new energy strategies (geothermal drillings, CO₂ storage, shell gas exploitations, bio fuel).
Water quality				
Water supply and sanitation		Mandatory measures to limit drinking water peak consumption during warm summer periods (e.g. "Phase orange", "Phase rouge").		
Extreme weather events				
Ecosystems				

1. The Flood warning service is operated in co-operation with the Water Management Agency and the Water navigation agency, www.inondations.lu (in French).

Main research programmes

- The Luxembourg Government supports research institutions to carry out climate change research related to vulnerability, mitigation and emission projections. - Luxembourg is currently participating in the regional Flow MS (Flood and Low Water Management Moselle-Sarre) project to assess the consequences of climate change for floods and low water flow in the Moselle and Saar catchments and to develop adjustment strategies.¹ - Adapting monitoring strategies in order to establish a long time series of surface and groundwater data to separate natural variability, human influences and climate change effects.

1. This project is supported by the Interreg IV-A program "Greater Region" (the Interreg Programme in the European Union: European Territorial Cooperation, commonly known as INTERREG, represents one of the pillars of this policy for the period 2007-13. It is financed by the European Regional Development Fund).

Principal financing mechanisms and investment programmes